PARMCO Hardware Parts List

ECSE 4235 Final Project — companion to Schematic Rev B (enable on GPIO 26)

Everything needed to build the circuit, grouped by job. The **Ref** column matches the labels on the schematic (J1, U1, R1…).

1. The Brain

Ref	Part	Qty	What it does (plain English)	Connects to
J1	**Raspberry Pi 4**	1	Runs your code and sends the control signals.	Everything; enable now on **GPIO 26 / Pin 37**

2. Motor Driving

Ref	Part	Qty	What it does (plain English)	Connects to
U2	**SN754410NE H-bridge (16-pin DIP)**	1	The muscle. Takes weak 3.3 V Pi signals and switches 12 V power into the motor coils.	Pi GPIO 18/23/24/25, motor, +5 V, +12 V, GND
M1	**NEMA 17 stepper motor (bipolar, 4-wire)**	1	The motor that turns. Two coils (A and B) energized in sequence make it step.	U2 pins 3, 6, 11, 14

3. Safety Switch (Enable Circuit)

Ref	Part	Qty	What it does (plain English)	Connects to
U1	**4N25 optocoupler (6-pin DIP)**	1	A light-powered switch. The Pi turns on a tiny LED inside, which turns the driver on. Keeps the Pi electrically separated from the driver's enable line.	Pin 1 ← R1 ← **GPIO 26**; Pin 4 → U2 pins 1 & 9
R1	**220 Ω resistor**	1	Limits current into the 4N25 LED so the Pi pin isn't overloaded.	**Pi Pin 37** → 4N25 Pin 1
R2	**10 kΩ resistor**	1	Pull-down: holds the driver OFF by default so the motor doesn't jump at boot.	4N25 Pin 4 → GND

4. Sensors

Ref	Part	Qty	What it does (plain English)	Connects to
U3	**AS5600 magnetic encoder**	1	Reads the motor shaft angle and reports it to the Pi over I²C.	3.3 V, GND, GPIO 2 (SDA), GPIO 3 (SCL); DIR → GND
U4	**IR speed sensor**	1	Counts how fast the shaft spins (2 pulses per revolution).	3.3 V, GND, GPIO 17

5. Power

Ref	Part	Qty	What it does (plain English)	Connects to
PS1	**Bench DC power supply (dual output)**	1	Supplies +5 V (driver logic) and +12 V (motor). Never powered from the Pi.	U2 pin 16 (+5 V), U2 pin 8 (+12 V), 4N25 pin 5 (+5 V), common GND

Remember: all grounds (Pi Pin 6, supply GND, U2 pins 4/5/12/13, 4N25 pin 2, sensor GNDs) must be tied together. You'll also need a breadboard and jumper wires to build it; those aren't listed in the project documents.